

Applications of Differential Calculus (Related Rates) Homework

① A 10 ft ladder rests against a wall. If the ladder slides away from the wall at 3 ft/s, how fast is the top of the ladder moving down the wall when the bottom of the ladder is 4 ft from the wall?

② A 13 m ladder rests against a wall. If the top of the ladder slides down the wall at $2\frac{1}{4}$ m/s, how fast is the bottom of the ladder moving away from the wall when the top of the ladder is 9 m from the ground?

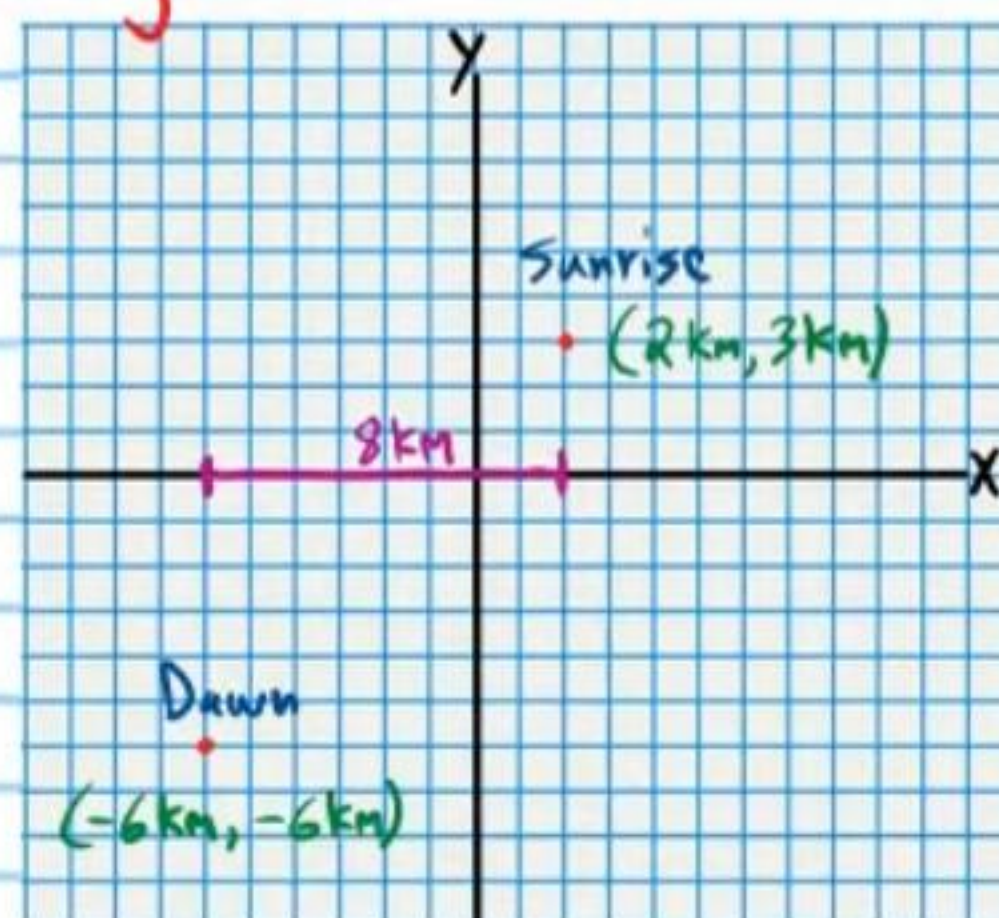
③ A camera at ground level films the lift-off of a rocket 800 yards from its launch pad. Find the rate of change in the angle of elevation of the camera when the rocket is moving at 20 yd/s and is 2000 yards above the ground. Assume that the camera is continually focused on the rocket.

④ Two helicopters are flying stationary at the same altitude and they are 400 meters apart. The helicopter at the left drops a rescue line to a man on the ground and reels him up while remaining stationary. When the man is 150 m from the helicopter rescuing him, the distance between the helicopter at the right and the man being rescued is changing by 10 m/min . At this moment in time, what is the rate of change of the angle of depression from the helicopter at the right to the man being rescued?

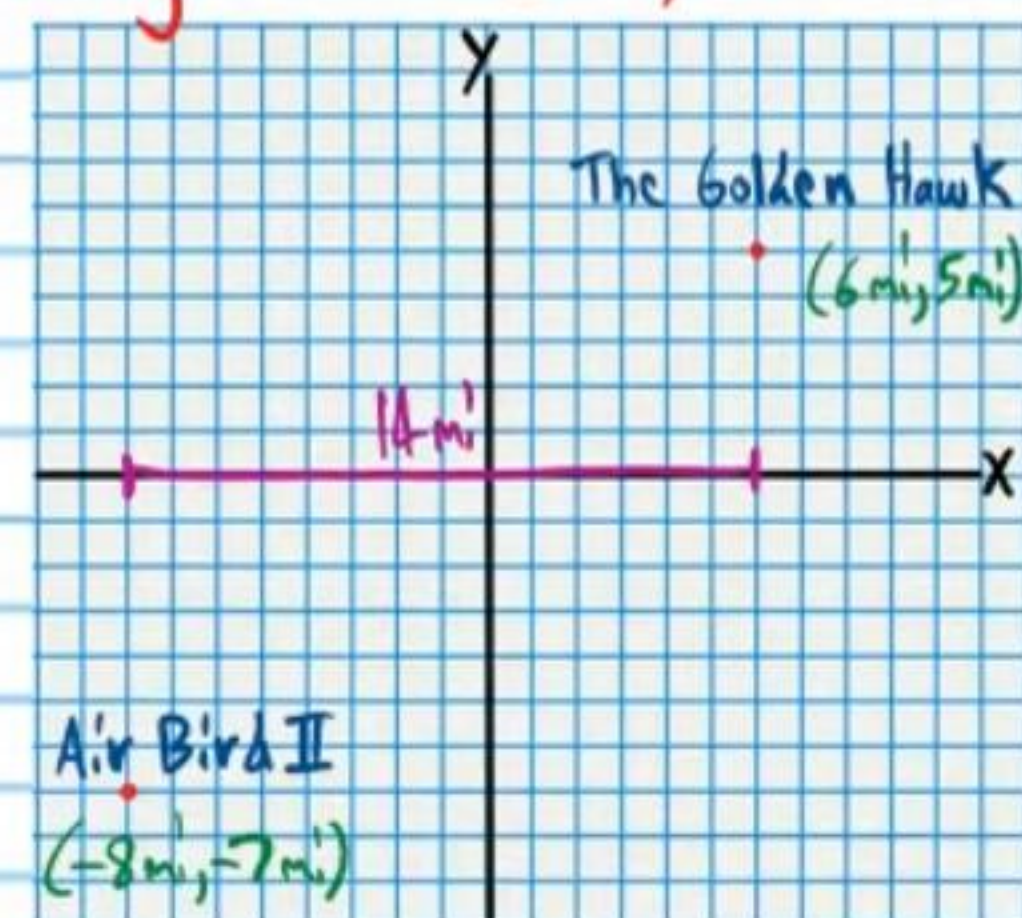
⑤ Two cars start to move from the same location. One travels north at 30 mi/h . The other travels east at 20 mi/h . What is the rate of change of the distance between the two cars 2 hours after they begin to move?

⑥ Larry is standing 100 ft north of Katie. They both begin to run at the same time. Larry runs east at 5 ft/s and Katie runs north at 8 ft/s . How fast is the distance between them changing 7 seconds after they begin to run? Warning! One of your rates will be negative.

⑦ Two ships are located at the coordinates below (each unit represents a kilometer). The "Sunrise" is moving south at 10 km/h and the "Dawn" is moving north at 15 km/h . How fast is the distance between them changing 10 minutes later? Warning! Your rates should be negative.



⑧ Two planes are located at the coordinates below (each unit represents a mile). The "Golden Hawk" is flying north at 8 mi/min and the "Air Bird II" is flying north at 5 mi/min . How fast is the distance between them changing one minute later? Warning! One of your rates should be negative. The altitude of each plane remains constant.



Homework Answers

① $-\frac{2\sqrt{21}}{7} \text{ ft/s}$ or -1.309 ft/s

② $\frac{81\sqrt{22}}{176} \text{ m/s}$ or 2.15866 m/s

③ $\frac{1}{290} \text{ rad/s}$ or 0.003448 rad/s

④ $-\frac{8\sqrt{73}}{1095} \text{ rad/s}$ or -0.06242 rad/s

⑤ $10\sqrt{13} \text{ mi/h}$ or 36.0555 mi/h

⑥ -3.148 ft/s

⑦ -10.8 km/h

⑧ 2.3158 m/min